Retailer Live Streaming Strategy in the Face of Manufacturer Encroachment Within Electronic Retailing

Lei Xie, Lu Dai, Pengwen Hou, and Pingping Chen

***Abstract*—As a novel effective format to enhance online sales, live streaming is increasingly popular in e-commerce retailing. Live streaming offers retailers a new strategy for dealing with manufacturer encroachment and has either a positive flow effect or a negative cannibalization effect on the possible direct channel. In this study, we construct a game-theoretic model to explore a retailer's live streaming strategy with a manufacturer who may encroach upon the retail market. We find that manufacturer encroachment always hurts the retailer in the absence of live streaming. However, in the presence of live streaming, the retailer can benefit from manufacturer encroachment if the flow effect of live streaming on the direct channel is significant. When the competition intensity is moderate and the flow effect is insignificant, with the manufacturer always encroaching, the retailer does not introduce live streaming even if the fixed cost is rather low; otherwise, the retailer will embrace live streaming at a lower fixed cost but not when it is high. Moreover, when the flow effect is significant, introducing live streaming when the fixed cost is low may motivate the manufacturer to choose encroachment even in the face of weak channel competition. Conversely, when the cannibalization effect of live streaming is significant, introducing live streaming when the fixed cost is low may discourage the manufacturer's encroachment when the competition intensity is moderate. However, in both scenarios, the manufacturer may benefit from the retailer's live streaming, resulting in a "win–win" situation, even if it prevents the manufacturer from encroaching.**

***Managerial Relevance Statement*—Our work enriches existing research on manufacturer encroachment and live streaming commerce by providing theoretical evidence that retailer live streaming can effectively respond to manufacturer encroachment. It also offers guidance for manufacturers' encroachment decisions, highlighting the importance of focusing on retailers' live streaming behavior and assessing factors such as channel competition intensity and potential spillover effects. Additionally, our work addresses retailer live streaming strategies, emphasizing the benefits of making live streaming decisions early and reducing fixed cost investments.**

***Index Terms*—Channel competition e-commerce, live streaming, manufacturer encroachment.**

I. Introduction

In THE past decade, the global electronic retail industry has experienced rapid development, with an average annual growth rate of up to 20% [1]. Data from research organizations indicates that the global e-commerce market reached approximately 5.7 trillion USD in 2022, accounting for 19.7% of the global retail sales. In the early stages of online retail platform development, retailers participated as distributors, purchasing products from manufacturers and reselling them to consumers. With the rapid development of online markets, some manufacturers have been establishing their own channels for selling products directly to consumers, in addition to indirectly selling products through retailers, a phenomenon typically referred to as "manufacturer encroachment" [2]. After implementing encroachment strategies, manufacturers gain direct channels to consumers. These channels, unaffected by double marginalization, typically enhance profit potential. Moreover, by making informed decisions about product pricing across both sales channels, manufacturers can improve control over product distribution and further maximize profitability. A considerable number of manufacturers employ such a multichannel approach. For example, Apple not only sells through the reselling channel of JD.com but also offers direct sales through its own official website. Similarly, Bose also sells through the reselling channel Amazon, and at the same time provides a direct channel.

As manufacturer encroachment competes for market share and enhances downstream competition, retailers have tried various tactics to respond to manufacturer encroachment [3], [4]. With the development of e-commerce technology, introducing live streaming has emerged as an effective way for retailers to enhance sales performance. In 2020, JD.com fully promoted the platform's self-owned stores to introduce live streaming, requiring the live streaming rate to reach 100%. In 2024, JD.com started a "purchase and sell" live room, specializing in wholesale products from manufacturers through official live streaming. In addition, other well-known online retail platforms, such as

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TikTok and Amazon, actively train professional live streaming teams or cooperate with well-known celebrities to introduce live streaming to promote sales. There is no doubt that retailer live streaming behavior might be a useful way to influence manufacturer encroachment because live streaming can enhance the experience of products or capitalize on the herd mentality of consumers, thereby increasing the market potential of the reselling channels to compete with manufacturers' direct channels.

In practice, retailer live streaming may have two opposite spillover effects on the manufacturer's direct channel. First, the retailer's live streaming behavior may have a "flow effect" on the direct channel, as consumers may get information about products through the retailer's live streaming and then turn to the direct channel, a phenomenon known as "live streaming free-riding". Second, the retailer's live streaming behavior may eat into the market potential of the direct channel, i.e., the live streaming attracts consumers who might have otherwise purchased through the direct channel to switch to the reselling channel, referred to as the "cannibalization effect" of live streaming. In addition, the introduction of live streaming will also incur corresponding costs, including one-off fixed costs such as equipment procurement and live streaming room set-up, as well as effort costs dedicated to enhancing the effectiveness of live streaming. The two opposed spillover effects, combined with the high cost of live streaming, complicate retailers' live streaming decision-making process. At the same time, these factors also cause the impact of retailers' live streaming on manufacturer encroachment to become more unpredictable.

Based on the aforementioned discussions, it is worth exploring the effect of retailers' live streaming behavior on manufacturers' channel encroachment strategies. Therefore, we are interested in the following three questions.

1) Whether and how retailer live streaming affects manufacturer encroachment? Can live streaming benefit retailers from manufacturer encroachment?
2) What is the incentive for retailers to introduce live streaming? How does the retailer's live streaming decision depend on factors such as the spillover effects of live streaming, live streaming costs, and competition intensity between two channels?
3) Does the retailer's live streaming decision change the manufacturer's encroachment decision? How does the retailer's live streaming decision affect the manufacturer?

To address the aforementioned issues, we construct a Stackelberg game with a manufacturer and a retailer. The retailer decides whether to introduce live streaming. Then, the manufacturer decides whether to encroach by adding a direct channel. Taking into account the retailer's adoption/nonadoption of live streaming and the manufacturer's encroachment/nonencroachment decisions, four cases are explored. By analyzing the optimal outcomes of each case, we derive some novel insights summarized as follows.

First, we find that the retailer's live streaming behavior does increase the complexity of manufacturer encroachment decisions and may counterintuitively make manufacturer encroachment favorable to the retailer. Specifically, in the absence of live streaming, the manufacturer typically chooses to encroach only when channel competition is exceptionally high, which inevitably hurts the retailer's profit. However, when the retailer introduces live streaming, the manufacturer may choose to encroach when live streaming exhibits a significant flow effect on the direct channel or the competition intensity is intense, and the retailer could benefit from manufacturer encroachment when the flow effect is significant.

Second, the retailer's live streaming strategy is closely related to the fixed cost of introducing live streaming, the spillover effects of live streaming, and the competition intensity between two channels. In general, the retailer introduces live streaming when the fixed cost of introducing live streaming is low. However, when the competition intensity is moderate and the flow effect is insignificant, with the manufacturer always encroaching, the retailer would not introduce live streaming even if the fixed cost is rather low.

Third, the retailer's live streaming decision indeed affects the manufacturer's encroachment decision. Specifically, when the flow effect is significant, the retailer's introduction of live streaming at a low fixed cost may induce the manufacturer to choose encroachment, even in the face of weak channel competition. However, when the cannibalization effect is significant and the competition intensity is moderate, the retailer's introduction of live streaming at a low fixed cost may instead inhibit the manufacturer's encroachment behavior. The manufacturer might benefit from the retailer's introduction of live streaming, creating a "win–win" situation, even if it prevents the manufacturer from encroaching.

The rest of this article is organized as follows. Section II reviews relevant studies. Section III constructs the model preliminaries and derives the retailer's and the manufacturer's optimal decision combinations. In Section IV, we analyze the strategic choices of the manufacturer and the retailer. In Section V, we extend our models by altering some assumptions in the basic model. Finally, Section VI concludes this article. All proofs are provided in the appendix.

II. Literature Review

Our work focuses on two areas: manufacturer encroachment and live streaming. In this section, we organize and summarize the research in these two areas.

A. Manufacturer Encroachment

Manufacturer encroachment has been a widely discussed issue in academia for many years [5], [6], [7], [8], [9]. There is a stream of work focused on retailers' anti-encroachment strategies. For instance, Huang et al. [3] found that retailers sharing demand information can deter manufacturers from encroachment. Liu and Zhang [4] indicated that both personalized pricing and manufacturer encroachment always hurt retailers, but retailers can use personalized pricing to discourage manufacturers from encroaching. Similarly, Zhang and Zhang [10] found that manufacturer encroachment always hurts retailers, yet retailers can expand their equity financing ratios to counter manufacturer encroachment. Liang et al. [11] argued that in the

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face of manufacturer encroachment, retailers have the option of switching suppliers and that this deterrent effect can discourage manufacturer encroachment. These articles show that retailers are harmed by manufacturer encroachment, but some strategies can effectively deter the manufacturer from encroaching.

In addition, several other articles have found that retailers can conditionally profit from manufacturer encroachment. For instance, Shi et al. [2] considered consumers' lower acceptance of online direct channels and found that brick-and-mortar retailers are likely to profit from manufacturer encroachment regardless of whether the manufacturer adopts an integrated or decentralized structure in the direct channel. Yoon [12] considered the investment spillovers and found that manufacturers implementing encroachment will make a selfish cost-reduction investment from which retailers may benefit. Liu et al. [13] examined manufacturer encroachment behavior with multiple retailers and found that if the manufacturer partnered with fewer retailers, then all retailers would benefit from manufacturer encroachment. Li et al. [14] and [15] found that retailers can profit from supplier encroachment when they have a demand information advantage. Dai et al. [16] considered the manufacturer's choice of agency or reselling encroachment at two retailers with asymmetric demand and found that setting a moderate slotting fee or commission rate enables the retailer to profit from manufacturer encroachment if the retailer's market potential is higher than another retailer. Li et al. [14], [15], and Dai et al. [16] have shown that even if manufacturer encroachment can't be prevented, there are strategies that retailers can adopt to profit from manufacturer encroachment.

The above research works either focus on retailers' antiencroachment strategies or get the bright side of manufacturer encroachment on retailers. Unlike the above studies, we focus on both the role of retailer live streaming on manufacturer encroachment and the impact of manufacturer encroachment on retailers who may introduce live streaming. In addition, to highlight the characteristics of retailer live streaming, we consider the spillover effects of live streaming on other channels, and such cross-channel effects of retailer strategies are also not considered in existing studies. Our results suggest that retailer live streaming not only can discourage manufacturer encroachment but even promotes manufacturer encroachment, allowing the retailer to profit from manufacturer encroachment.

B. Live Streaming

Our work is also closely related to live streaming commerce. Existing research was mostly empirical, focusing on the factors that influence consumer behavior in live streaming, such as utilitarian and hedonic values [17], viewer's perceived value [18], [19], streamer sentiment [20], [21], and technological factors of live streaming [22]. Several literatures have used theoretical modeling approaches to study pricing or quality decisions [23], [24], selling model selection [25], [26], [27], live streaming models [28], and streamer or manufacturer selection [29], [30]. Several studies above have considered the cannibalization effect of live streaming channels on other channels. For instance, Wang et al. [27] studied the options of reselling and agency sales between suppliers and online platforms, considering the cannibalization effect of live streaming channels on traditional online channels. Similarly, Zhang et al. [28] examined manufacturers' choices between merchant live streaming and influencer live streaming, considering the different cannibalization effects of different live streaming modes on traditional online channels.

In addition, some studies focused on the live streaming strategy and its impact on supply chain members. For instance, Pan et al. [31] considered consumer demand switching and found that the introduction of live streaming increases firm profits only when the streamer's selling power is high. Wang and Zhang [32] found that manufacturers should distribute products through the influencer channel when the potential of misfit elimination through the influencer channel is not too high and the hassle cost is relatively low. Zhang et al. [33] discussed whether multinational enterprises should establish live streaming on overseas e-commerce platforms and found that the introduction of live streaming channels always benefits multinational enterprises, but harms the profits of overseas e-tailing and third-party e-retailers. Wang et al. [34] developed two influencer live streaming models under the revenue-sharing contract and the flow-based contract to study the conditions under which manufacturers should introduce live streaming services, considering the flow effect of live streaming channels on traditional online channels.

Our study differs from existing research above in two main aspects. First, differing from Wang et al. [27], Zhang et al. [28], and Wang et al. [34], we consider both the flow effect and cannibalization effect of live streaming channels on traditional online channels, not just one or the other. Second, we combine retailer live streaming with manufacturer encroachment to explore their interactions, especially the impact of retailer live streaming on the manufacturer's encroachment strategy, not just the impact of live streaming on supply chain members. Our results suggest that when the competition intensity is moderate and the live streaming cannot bring a significant flow effect on the direct channel, with the manufacturer always encroaching, the retailer should not introduce live streaming even if the live streaming fixed cost is rather low. In addition, the manufacturer might benefit from the retailer live streaming, even if it prevents the manufacturer from encroaching.Table I concludes and compares the most relevant literature. Our study makes three contributions. First, to the best of our knowledge, our study is the first to combine manufacturer encroachment and retailer live streaming introduction to explore their interactions. Second, we capture the characteristics of live streaming commerce while considering the possible positive flow effect and negative cannibalisation effect of live streaming. Third, we draw counterintuitive conclusions, e.g., we find that the manufacturer tends to profit from retailer live streaming even if it prevents manufacturer encroachment, and that retailers should not introduce live streaming even if they invest a greatly low fixed cost. These findings provide important insights for manufacturers' optimization of channel decisions and retailers' live streaming introduction decisions in practice.

III. Model and Equilibrium Results

Consider a supply chain with a manufacturer (for brevity, referred to as m or he) wholesales products at unit wholesale

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TABLE I
COMPARISON BETWEEN THIS ARTICLE AND OTHER RELATED LITERATURE

Literature	Manufacturer encroachment	Anti-encroachment	Live streaming introduction	Flow effect	Cannibalization effect
Liu and Zhang [4]	✓	✓			
Zhang and Zhang [10]	✓	✓			
Liang *et al.* [11]	✓	✓			
Huang *et al.* [3]	✓	✓			
Yoon [12]	✓	✓			
Shi *et al.* [2]	✓				
Liu *et al.* [13]	✓				
Li *et al.* [14]	✓				
Li *et al.* [15]	✓				
Dai *et al.* [16]	✓				
Pan *et al.* [31]			✓		
Wang and Zhang [32]			✓		
Zhang *et al.* [33]			✓		
Wang *et al.* [27]					✓
Zhang *et al.* [28]					✓
Wang *et al.* [34]			✓	✓	
This paper	✓	✓	✓	✓	✓

price w to a retailer (for brevity, referred to as r or she) that operates a reselling channel, and the retailer sells products to consumers at unit retail price p_r. In addition, the manufacturer also has the option to establish a direct channel, enabling him/her to sell products directly to consumers. With the development of the Internet and cloud computing technology, the cost of website construction and maintenance required for online direct channels has been significantly reduced. For example, Shopify provides a one-stop solution from website building to payment, and the powerful open-source e-commerce platform Magento provides free website-building functions. Therefore, following the approach of Yoon [12], we assume that the channel construction cost is 0 to focus on the impact of other main factors, such as competition intensity, on the manufacturer's encroachment strategy. Besides, we assume that the manufacturer and retailer are both risk-neutral and aim to maximize profits [35], [36]. Table II describes the variables.

Following the setup in Liu et al. [24], we consider that the retailer could introduce live streaming with effort level e to improve operational effectiveness and deal with the threat of encroachment by the manufacturer.[1] This effort may be reflected in gaining attention and purchasing capacities through the streamer's real-time social interactions and illustrations with customers. The retailer introducing live streaming can expand the retail channel's market size to $a + e$ [24], where a represents baseline market size. With the potential market size, we assume the consumer demand follows a linear, downward-sloping function [1], [13], [37]. Without manufacturer encroachment, the demand function for the reselling channel is represented as

$$q_r = a - p_r + \mathbb{I}e, \tag{1}$$

where q_r and p_r are, respectively, the retail price and demand in the reselling channel, $\mathbb{I} = 1$ or 0 is the indicator for the retailer's decision with or without live streaming.

If the manufacturer encroaches, a competition relationship is formed between the manufacturer and the retailer in the market, and live streaming introduction generates further cross-channel spillover of live streaming effort. They compete by adjusting the retail price and the live streaming effort level. The demand

TABLE II
DESCRIPTION OF PARAMETERS AND VARIABLES

Parameters	Description
a	Total market size.
m	Market share of the direct channel.
b	Competition intensity between the two channels, $b \in (0, 1)$.
θ	Spillover factor of live streaming effort, $\theta \in (-1, \hat{\theta})$.
c_r	Fixed cost of introducing live streaming.
c_l	Variable cost of introducing live streaming, $c_l = \frac{e^2}{2}$.
Decision variables	**Description**
e	Live streaming effort level.
w	Wholesale price of the product.
p_m	Direct price of the product in the direct channel.
p_r	Retail price of the product in the reselling channel.
Calculated variables	**Description**
q_m	Demand for the product in the direct channel.
q_r	Demand for the product in the reselling channel.
π_m	Profit of the manufacturer.
π_r	Profit of the retailer.

[1]To focus on the impact of retailer live streaming on manufacturer encroachment, we only consider retailer live streaming rather than manufacturer live streaming. In practice, many manufacturers, such as SL Bearing and Leyard, have not yet integrated live streaming into their sales strategies.

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functions for the direct channel and reselling channel are

$$\begin{cases} q_m = ma - p_m + bp_r + \mathbb{I}\theta e \\ q_r = (1-m)a - p_r + bp_m + \mathbb{I}e, \end{cases} \tag{2}$$

where p_m and q_m represent the direct price and demand in the direct channel. Parameter m captures the market share of the direct channel. In other words, with manufacturer encroachment, the baseline market size for the manufacturer's direct channel is ma, and the baseline market size for the reselling channel is $(1-m)a$. We assume that the two channels share the underlying market equally, i.e., $m = \frac{1}{2}$, to focus on the effect of key factors such as channel competition intensity and spillover factor. Additionally, in Section V-A, we relax this assumption and explore its impact on live streaming and encroachment strategies to verify the robustness of our main conclusions. Parameter $b \in (0,1)$ denotes the competition intensity between different channels in the supply chain. A higher b indicates more intense competition between the direct channel and the reselling channel. Following the setup in Wang et al. [27], Zhang et al. [28], Wang et al. [34], parameter θ captures the cross-channel spillover effects of the live streaming effort between the two channels, which may be a negative cannibalization effect ($\theta < 0$) or a positive flow effect ($\theta > 0$). Since the live streaming effort affects its own channel more than another channel, we assume $\theta \in (-1, \hat{\theta})$, $0 < \hat{\theta} < 1$, and guarantee all the equilibrium outcomes are positive. A larger spillover factor θ indicates that the retailer's live streaming effort is more favorable to the potential demand of the direct channel.

If the retailer introduces live streaming, she incurs two types of costs. First, there is a fixed cost associated with establishing the live streaming channel, denoted as c_r. The fixed cost includes expenses for training live streaming personnel and equipment acquisition, and is a one-time payment during the initial setup of the live streaming channel, independent of the live streaming effort level or the quantity of product sold. Second, there is a variable cost associated with the live streaming effort level e, denoted as c_l. We assume that $c_l = \frac{e^2}{2}$. This convex and increasing form of the effort cost is widely applied in the previous operations management studies on live streaming [24], [28], [38]. This cost is convex increasing with respect to e determined by the retailer linked in the live streaming practice. In such a setting, when the retailer attempts to improve live streaming effort, she initially reaps "low-hanging fruit" but then must make a larger cost to secure further improvement.

With the expressions of demand and cost, we derive the profit functions of the manufacturer and the retailer. When the manufacturer does not encroach, his profit solely comes from the wholesale revenue in the reselling channel. Correspondingly, when the manufacturer encroaches, his profit consists of the wholesale revenue in the reselling channel and the sales revenue in the direct channel. The retailer's profit comes from the net profit of the retail income minus the wholesale expenditure in the reselling channel. If the retailer introduces live streaming, she also needs to pay the costs of live streaming, including fixed cost c_r and variable cost c_l. Thus, the manufacturer's profit function is

$$\pi_m = \begin{cases} wq_r & \text{without encroachment} \\ wq_r + p_m q_m & \text{with encroachment}, \end{cases} \tag{3}$$

and the retailer's profit function is

$$\pi_r = (p_r - w)q_r - \mathbb{I}(c_r + c_l). \tag{4}$$

To focus on the impact of the retailer's live streaming strategy on manufacturer encroachment, we assume that the retailer's live streaming decision is made before manufacturer encroachment, and in Section V-B, we consider the retailer's live streaming decision to be later than manufacturer encroachment to verify the robustness of the findings. Fig. 1 illustrates the sequence of events in our game model. First, the retailer decides whether to introduce live streaming. Second, the manufacturer decides whether to encroach. Third, the manufacturer sets the wholesale price w. Fourth, the retailer sets the retail price p_r, and if the manufacturer chooses to encroach, he sets the direct price p_m. Fifth, if the retailer introduces live streaming, she sets the live streaming effort level e. In the end, demands are realized, and the participants generate profits.

We use E(N) and L(W) to denote the manufacturer's encroachment decision and the retailer's live streaming decision, respectively. Here, E(N) denotes encroachment (no encroachment) and L(W) denotes live streaming (without live streaming). The different encroachment decisions of the manufacturer or live streaming decisions of the retailer create four possible cases: (a) case NW, the manufacturer does not encroach, and the retailer does not introduce live streaming; (b) case EW, the manufacturer encroaches, and the retailer does not introduce live streaming; (c) case NL, the manufacturer does not encroach, and the retailer introduces live streaming; (d) case EL, the manufacturer encroaches, and the retailer introduces live streaming.

Based on the description above, this article models the game processes between the manufacturer and retailer in four different cases and obtains equilibrium solutions as shown in Table III. The specific solving process is provided in the appendix.

IV. Analysis

In the previous section, we calculated the optimal decisions and corresponding equilibrium states for the manufacturer and retailer under different cases. Next, we will investigate the manufacturer's encroachment strategy and the retailer's preference for manufacturer encroachment, as demonstrated in Sections IV-A and IV-B, respectively, and then examine the retailer's live streaming strategy and the impact of the retailer's live streaming on the manufacturer, as shown in Sections IV-C and IV-D, respectively.

A. Manufacturer Encroachment Decision

1) Without Retailer Live Streaming: We first study whether the manufacturer should encroach given that the retailer does not introduce live streaming. The manufacturer's optimal encroachment decision occurs by comparing his profits under encroachment and nonencroachment. Then, we characterize the manufacturer's optimal encroachment decision in the absence of the retailer's live streaming as follows.

Proposition 1: Given that the retailer does not introduce live streaming, the manufacturer does not encroach if $0 < b < b_1$; otherwise, the manufacturer encroaches.

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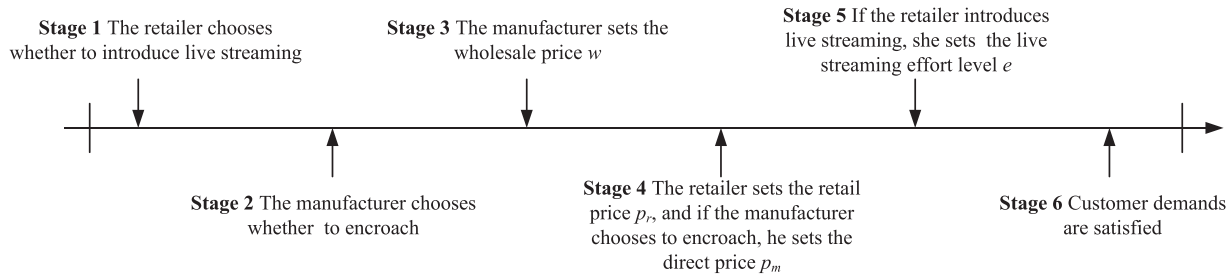


Fig. 1. Game sequence.

TABLE III
EQUILIBRIUM OUTCOMES IN FOUR CASES

	NW	EW	NL	EL
w	$\frac{a}{2}$	$\frac{a(b^3+8)}{4(8-b^3+b^2-8b)}$	$\frac{a}{2}$	$\frac{a\left(b^4+2\,b^3\theta+b^2\theta^2-2\,b^2-4\,b\theta-2\,\theta^2+2\,b-2\,\theta+4\right)}{4(b-1)(b+1)(b+\theta+2)(b+\theta-2)}$
p_r	$\frac{3a}{4}$	$\frac{a(b^3-2b^2+4b-12)}{4(b^3-b^2+8b-8)}$	a	$\frac{a\left(b^4-b^2\theta^2+2\,b^3+2\,b^2\theta+4\,b^2+2\,b\theta+2\,\theta^2-4\,b-8\right)}{4(1-b)(b+1)(b+\theta+2)(b+\theta-2)}$
p_m	–	$\frac{a(b-4)(b+2)}{4(b^3-b^2+8b-8)}$	–	$\frac{a(3\,b+\theta+2)\left(b^2+b\theta-2\right)}{4(1-b)(b+1)(b+\theta+2)(b+\theta-2)}$
e	–	–	$\frac{a}{2}$	$\frac{a\left(b^2+b\theta+b+\theta+2\right)}{2(b+\theta+2)(2-b-\theta)}$
π_m	$\frac{a^2}{8}$	$\frac{a^2(b+2)\left(b^2-b+6\right)}{16\,(1-b)(b^2+8)}$	$\frac{a^2}{4}$	$\frac{a^2\left(b^4+2\,b^3\theta+b^2\theta^2+4\,b^2+4\,b\theta+12\,b+4\,\theta+8\right)}{16\,(b-1)(b+1)(b+\theta+2)(b+\theta-2)}$
π_r	$\frac{a^2}{16}$	$\frac{a^2\left(b^2+2\right)^2}{4\,(b^2+8)^2}$	$\frac{a^2}{8}-c_r$	$\frac{a^2\left(b^2+b\theta+b+\theta+2\right)^2}{8\,(b^2+2\,b\theta+\theta^2-4)^2}-c_r$

Proposition 1 reveals that encroachment arises when the competition between the reselling and direct channel is intense ($b_1 < b < 1$). This consequence runs counter to the prevalent belief that the manufacturer may not have an incentive to encroach on the market amidst fierce competition. However, the finding is reasonable because when the manufacturer encroaches, a relationship with coexisting competition and cooperation is formed between the manufacturer and retailer. In this context, the manufacturer is more likely to raise the wholesale price, which forces the retailer to raise the retail price, thereby reducing demand in the reselling channel. The manufacturer shifts demand to his own direct channel by raising the wholesale price and gains a dominant position in the market. Intense competition reinforces the manufacturer's dominant position. More consumers turn to the manufacturer's direct channel, which increases the manufacturer's profit. Thus, when the competition intensity reaches a certain level, the manufacturer is more inclined to encroach, whereas such behavior is avoided when competition is relatively less intense.

2) With Retailer Live Streaming: First, we explore the impact of manufacturer encroachment on the retailer's live streaming effort decision.

Proposition 2: Given that the retailer introduces live streaming, the retailer reduces live streaming effort when the manufacturer encroaches if $-1 < \theta < \theta_1$; otherwise, the retailer raises live streaming effort when the manufacturer encroaches.

Proposition 2 shows that manufacturer encroachment affects the retailer's live streaming effort decision because of the spillover of live streaming effort (i.e., θ). When the spillover factor is small (i.e., $-1 < \theta < \theta_1$), if the manufacturer implements encroachment, in order to mitigate the cross-channel spillover of live streaming effort, he will increase the wholesale price, thereby increasing the retailer's purchasing cost. This forces the retailer to reduce investment in the live streaming effort to cope with the higher cost pressure. Conversely, when the spillover factor is large (i.e., $\theta_1 < \theta < \hat{\theta}$), as live streaming effort becomes more favorable to the direct channel, the manufacturer will lower the wholesale price to induce the retailer to increase live streaming effort. Thereby, the manufacturer's optimal encroachment strategy with retailer live streaming is characterized as follows.

Proposition 3: Given that the retailer introduces live streaming, the manufacturer does not encroach if $0 < b < b_2$ and $-1 < \theta < \theta_2$; otherwise, the manufacturer encroaches.

Proposition 3 demonstrates that if the retailer introduces live streaming, the manufacturer's optimal encroachment decision depends not only on the channel competition between the direct and reselling channel (i.e., b) but also on the spillover effects of the live streaming effort (i.e., θ). Specifically, only when the channel competition is small ($0 < b < b_2$) and the spillover factor is small ($-1 < \theta < \theta_2$), the manufacturer will not encroach; otherwise, encroachment is profitable. Manufacturer demand shifting is ineffective when competitive intensity is low. If the spillover factor is small, according to Proposition 2, manufacturer encroachment causes the retailer to reduce live streaming effort, which mitigates the spillover on the direct

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channel but drastically cuts the demand of the reselling channel. Thus, inefficient demand shifting combined with a small spillover factor makes manufacturer encroachment unprofitable, and manufacturer encroachment is profitable only when a larger spillover factor dominates inefficient demand shifting caused by the low competition intensity. Furthermore, when the competition intensity is intense ($b_2 < b < 1$), the manufacturer has a strong ability to shift demand between the two channels by setting the wholesale price. Regardless of whether the spillover of the live streaming effort is a positive flow effect or a negative cannibalization effect, the manufacturer could profit from encroachment.

B. Retailer's Preference for Manufacturer Encroachment

One might anticipate that the intensified competition between the manufacturer and the retailer, resulting from the manufacturer's encroachment, typically has a detrimental impact on the retailer's profitability. However, it is not clear whether this argument still holds when the retailer has an option to introduce live streaming, especially when considering the spillover effects of live streaming effort. To address this question, we examine the retailer's preferences for the manufacturer's encroachment strategy without and with live streaming, respectively, which are characterized as follows.

Proposition 4: The impact of the manufacturer encroachment strategy on the retailer's profit.

i) Without retailer live streaming, manufacturer encroachment always hurts the retailer.
ii) With retailer live streaming, manufacturer encroachment benefits the retailer if $\theta_1 < \theta < \hat{\theta}$; otherwise, manufacturer encroachment hurts the retailer.

In Proposition 4, we can assess the impact of the manufacturer encroachment on the retailer's profitability. In the absence of live streaming, as shown in Proposition 4 (i), the result is consistent with conventional wisdom, wherein the manufacturer encroachment always hurts the retailer's profitability. When the manufacturer chooses to encroach, he enters into competition with the retailer in the market. This leads to a portion of consumers shifting their purchases to the manufacturer's direct channel (as we set $m = \frac{1}{2}$, earlier). Moreover, as the leader in the supply chain, the manufacturer can adjust not only the wholesale price supplied to the retailer but also the direct price to compete with the retailer. This further weakens the retailer's competitive position in the supply chain. Taking into account the above two factors, the retailer always prefers the manufacturer not to encroach if the retailer does not introduce live streaming.

Proposition 4 (ii) shows that with the retailer introducing live streaming, manufacturer encroachment may benefit the retailer. This situation occurs when the spillover factor is large (i.e., $\theta_1 < \theta < \hat{\theta}$). This consequence virtually runs counter to the prevalent belief that manufacturer encroachment always hurts the retailer (see, e.g., Liu and Zhang [4], Frazier and Lassar [39], Fein and Anderson [40]). We argue the reason that although the manufacturer encroachment behavior leads to a redistribution of market share from the retailer, the retailer is incentivized to invest in a higher live streaming effort to attract more consumers, especially when the spillover factor is large, as mentioned in Proposition 2. Therefore, the retailer may prefer the manufacturer to encroach if she introduces live streaming. This indicates that the retailer should not always oppose the manufacturer's encroachment. When the live streaming effort has a positive flow effect on the direct channel, the retailer may even derive benefits from the manufacturer encroachment because of the lower wholesale price.

TABLE IV
LIVE STREAMING AND ENCROACHMENT EQUILIBRIUM

b	θ	c_r	Retailer	Equilibrium strategy
$0 < b < b_1$	$-1 < \theta < \theta_2$	$c_r < c_{r1}$	Live streaming	NL
		$c_r > c_{r1}$	Without live streaming	NW
	$\theta_2 < \theta < \hat{\theta}$	$c_r < c_{r2}$	Live streaming	EL
		$c_r > c_{r2}$	Without live streaming	NW
$b_1 < b < b_3$	$-1 < \theta < \theta_2$	$c_r < c_{r3}$	Live streaming	NL
		$c_r > c_{r3}$	Without live streaming	EW
	$\theta_2 < \theta < \hat{\theta}$	$c_r < c_{r4}$	Live streaming	EL
		$c_r > c_{r4}$	Without live streaming	EW
$b_3 < b < b_2$	$-1 < \theta < \theta_2$	$c_r < c_{r3}$	Live streaming	NL
		$c_r > c_{r3}$	Without live streaming	EW
	$\theta_2 < \theta < \theta_3$	$\forall c_r$	Without live streaming	EW
	$\theta_3 < \theta < \hat{\theta}$	$c_r < c_{r4}$	Live streaming	EL
		$c_r > c_{r4}$	Without live streaming	EW
$b_2 < b < b_4$	$-1 < \theta < \theta_3$	$\forall c_r$	Without live streaming	EW
	$\theta_4 < \theta < \hat{\theta}$	$c_r < c_{r4}$	Live streaming	EL
		$c_r > c_{r4}$	Without live streaming	EW
$b_4 < b < 1$	$\forall\theta$	$c_r < c_{r4}$	Live streaming	EL
		$c_r > c_{r4}$	Without live streaming	EW

C. Retailer Live Streaming Decision

Building upon the manufacturer's optimal encroachment decision, we move to the first stage to examine the retailer's optimal live streaming decision. When the retailer does not introduce live streaming, her profit conditional on the manufacturer's encroachment decision is

$$\pi_r^{\mathrm{W}} = \begin{cases} \frac{a^2}{16} & \text{if } 0 < b < b_1 \\ \frac{a^2\left(b^2+2\right)^2}{4\left(b^2+8\right)^2} & \text{if } b_1 < b < 1. \end{cases} \tag{5}$$

Otherwise, the retailer's profit when she introduces live streaming is represented by

$$\pi_r^{\mathrm{L}} = \begin{cases} \frac{a^2}{8} - c_r \text{if } 0 < b < b_2 \text{ and } -1 < \theta < \theta_2 \\ \frac{a^2\left(b^2+b\theta+b+\theta+2\right)^2}{8\left(b^2+2\,b\theta+\theta^2-4\right)^2} - c_r \text{ if } 0 < b < b_2 \text{ and} \\ \theta_2 < \theta < \hat{\theta}, \text{ or } b_2 < b < 1. \end{cases} \tag{6}$$

The retailer's live streaming decision and the equilibrium strategy are summarized in Proposition 5 and Table IV as follows.

Proposition 5: When $b_3 < b < b_2$ and $\theta_2 < \theta < \theta_3$, or $b_2 < b < b_4$ and $-1 < \theta < \theta_3$ $(\theta_3 < 0)$, the retailer does not introduce live streaming no matter how small the fixed cost c_r, otherwise, the retailer does not introduce live streaming if $c_r > c_{ri}$ and introduces live streaming if $c_r < c_{ri}$, $i \in \{1, 2, 3, 4\}$.

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Proposition 5 states the retailer's optimal live streaming strategy building upon the manufacturer's optimal encroachment decision as established in Propositions 1 and 3 and shown in Table IV. Note that when the retailer is faced with a tradeoff between EW and EL (i.e., $b_1 < b < b_2$ and $\theta_2 < \theta < \hat{\theta}$, or $b_2 < b < 1$), the retailer does not introduce live streaming no matter how small the fixed cost c_r if the competition intensity is moderate and the spillover factor is not large enough (i.e., $b_3 < b < b_2$ and $\theta_2 < \theta < \theta_3$, or $b_2 < b < b_4$ and $-1 < \theta < \theta_3$). At this point, the manufacturer may significantly increase the wholesale price to curb the retailer's investment in live streaming effort, the higher wholesale cost and live streaming variable cost make introducing live streaming always unprofitable for the retailer. Otherwise, the retailer does not introduce live streaming if $c_r > c_{r4}$ and introduces live streaming if $c_r < c_{r4}$. Similarly, there exists three thresholds under the other three scenarios, $c_{ri}, i \in \{1, 2, 3\}$, with respect to the fixed cost of introducing live streaming. Intuitively, when the fixed cost of introducing live streaming is lower than this threshold, the retailer enjoys a large efficiency advantage in the live streaming selling process, which helps her to gain more profit. In contrast, when the fixed cost is too large, the retailer is made worse off by investing a significant cost to introduce live streaming. Additionally, as we can see from Table IV, the retailer's live streaming may act as an effective method for affecting the manufacturer's encroachment behavior, which is summarized in the following proposition.

Proposition 6: The impact of the retailer's live streaming decision on the manufacturer's encroachment decision is summarized as follows.

i) When $0 < b < b_1$ and $\theta_2 < \theta < \hat{\theta}$, the retailer introduces live streaming and prompts the manufacturer to encroach if $c_r < c_{r2}$.
ii) When $b_1 < b < b_2$ and $-1 < \theta < \theta_2$, the retailer introduces live streaming and discourages the manufacturer from encroaching if $c_r < c_{r3}$.

Proposition 6 reveals the impact of retailer's live streaming on manufacturer encroachment. Under certain conditions, the retailer's live streaming behavior does affect the manufacturer's encroachment. This result is seemingly in accord with those of most extant studies (see, e.g., Huang et al. [3], Liu and Zhang [4], Zhang and Zhang [10]). However, we find that retailer live streaming has opposite effects on manufacturer encroachment under different conditions, not just countering encroachment. Specifically, Proposition 6 (i) states that when competition intensity is weak (i.e., $0 < b < b_1$) and the spillover factor is large (i.e., $\theta_2 < \theta < \hat{\theta}$), the retailer introduces live streaming if the fixed cost is low (i.e., $c < c_{r2}$), and prompts the manufacturer to encroach. We argue that the reason is as follows: when the fixed cost of live streaming is high, the retailer does not live streaming, and the manufacturer does not encroach due to the limited demand shifting capacity resulting from the weak competitive intensity. However, when the retailer introduces live streaming at a low fixed cost, manufacturer encroachment can motivate the retailer to raise live streaming effort, especially when the spillover factor is large. The retailer's increased live streaming effort not only attracts more consumers to buy the product in the reselling channel but also is favorable to the direct channel, both of which increase the manufacturer's profit. Therefore, the retailer's live streaming behavior will contribute to the manufacturer's encroachment.

Proposition 6 (ii) states that when the competition intensity is moderate (i.e., $b_1 < b < b_2$) and the spillover factor is small (i.e., $-1 < \theta < \theta_2$), the retailer introduces live streaming if the fixed cost is low (i.e., $c < c_{r3}$), and discourages the manufacturer from encroaching. According to Proposition 1, if the retailer does not introduce live streaming when the fixed cost is large, the manufacturer encroaches when the competition intensity is moderate. However, when the retailer introduces live streaming at a low fixed cost, the manufacturer's own weaker demand-shifting ability, and the lower live streaming effort caused by the small spillover factor, make encroachment unprofitable for the manufacturer. Therefore, in this situation, the retailer's live streaming behavior will discourage the manufacturer's encroachment. To further illustrate those intriguing findings, we depict the retailer's profits with and without live streaming for different θ and b, as illustrated in Fig. 2. Thereinto, the dashed and solid lines depict the retailer's profits with and without live streaming, respectively; "EW (NW)" denotes the manufacturer's "encroachment (no encroachment)" without retailer live streaming, and "EL (NL)" signifies the manufacturer's "encroachment (no encroachment)" with retailer live streaming. Note that in order to focus on the impact of the live streaming on the manufacturer's encroachment strategy, we only despite figures for which live streaming channels are less costly to build (i.e., $c_r = 0.3$, satisfying $c_r < c_{ri}, i \in \{1, 2, 3, 4\}$). With Fig. 2(a), we find that when the competition intensity is small, if the spillover factor is large enough, the retailer's live streaming at a low fix cost induces the manufacturer to encroach, resulting equilibrium strategy shifts from "NW" to "EL", as shown in Proposition 6 (i). Similarly, by combining Fig. 2(b) with Proposition 6 (ii), we find that when the competition intensity is moderate and the spillover factor is small, the retailer's live streaming behavior at a low fixed cost will discourage the manufacturer from encroaching, resulting equilibrium strategy shifts from "EW" to "NL".

D. Impact of Retailer Live Streaming on the Manufacturer

Building upon the manufacturer's optimal encroachment decision under different live streaming strategies, next, we turn to explore the impact of the retailer's live streaming on the manufacturer and summarize the result in the following proposition.

Proposition 7: When $b_5 < b < 1$ and $-1 < \theta < \theta_4$, retailer live streaming hurts the manufacturer; otherwise, retailer live streaming benefits the manufacturer even if it discourages the manufacturer from encroaching.

Proposition 7 presents the impact of the retailer's live streaming strategy on the manufacturer. Specifically, when the competition is intense (i.e., $b_5 < b < 1$) and the spillover factor is small (i.e., $-1 < \theta < \theta_4$), the manufacturer prefers the retailer to avoid introducing live streaming because it requires the manufacturer to raise the wholesale price significantly to dampen the live streaming effort investment, which in turn mitigates the impact of live streaming on the direct channel, as the region enclosed by the dotted line in Fig. 3 shows. Note

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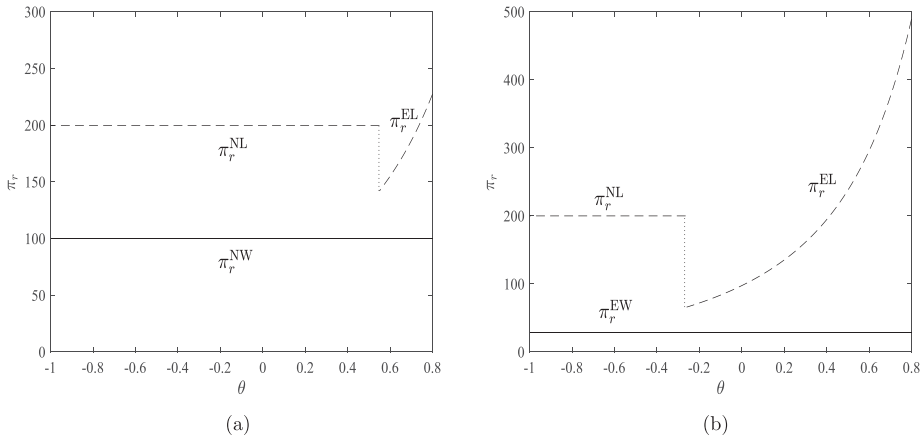


Fig. 2. Retailer's profits with and without live streaming ($a = 40$, $c_r = 0.3$). (a) $b = 0.2$. (b) $b = 0.45$.

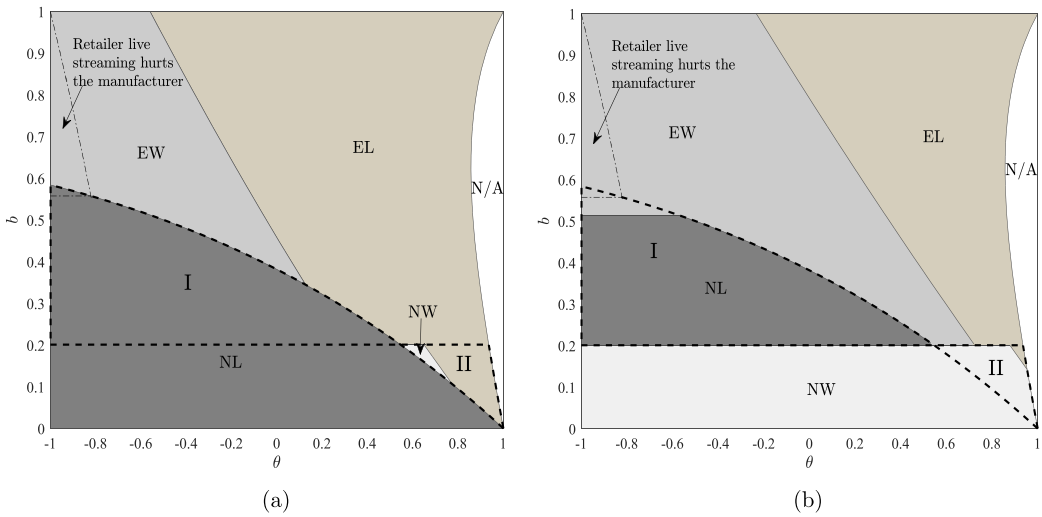


Fig. 3. Retailer's live streaming decision and impact on the manufacturer ($a = 40$). (a) $c_r = 70$. (b) $c_r = 170$.

that the retailer may similarly be reluctant to introduce live streaming, which unexpectedly prompts a "win–win" situation between the manufacturer and the retailer in "EW". Except for the above region, the manufacturer always benefits from the retailer's live streaming. The region I enclosed by the dashed line in Fig. 3 indicates that the retailer faces a tradeoff between "EW" and "NL" when the competition intensity is moderate and the spillover factor is small, as stated in Proposition 5 (ii), the retailer live streaming at a low fixed cost prevents the manufacturer from encroachment. Besides, the manufacturer might also prefer the retailer live streaming, which prompts a "win–win" situation between the manufacturer and the retailer in "NL", even if retailer live streaming prevents him/her from encroaching. Furthermore, region II enclosed by the dashed line indicates that the retailer faces a tradeoff between "NW" and "EL" when the competition is small and the spillover factor is large. As stated in Proposition 5 (i), the retailer's live streaming at a low fixed cost induces the manufacturer to encroach, and the manufacturer benefits from both the retailer's live streaming and the encroachment, thereby two parties realize a "win–win" situation of "EL". Moreover, combining Fig. 3(a) and (b), we find that when the fixed cost of introducing live streaming is lower, not only the area in which the retailer introduces live streaming expands but also the area in which both parties realize a "win–win" situation of live streaming expands.

In summary of our main findings from Propositions 5–7 and fig. 3, for the retailer, lowering the fixed cost of introducing live streaming, in other words, investing a lower fixed cost when choosing to introduce live streaming, not only influences the decision to encroach by the manufacturer but also leads to a "win–win" situation with the manufacturer, even if the manufacturer chooses to encroach.

V. Extensions

In this section, we extend a set of assumptions made in the basic model to generalize further insights. The proofs are provided in the Appendix.

A. Different Market Shares

Thus far we have assumed that the direct channel has the same market share as the reselling channel, i.e., $m = \frac{1}{2}$. In this

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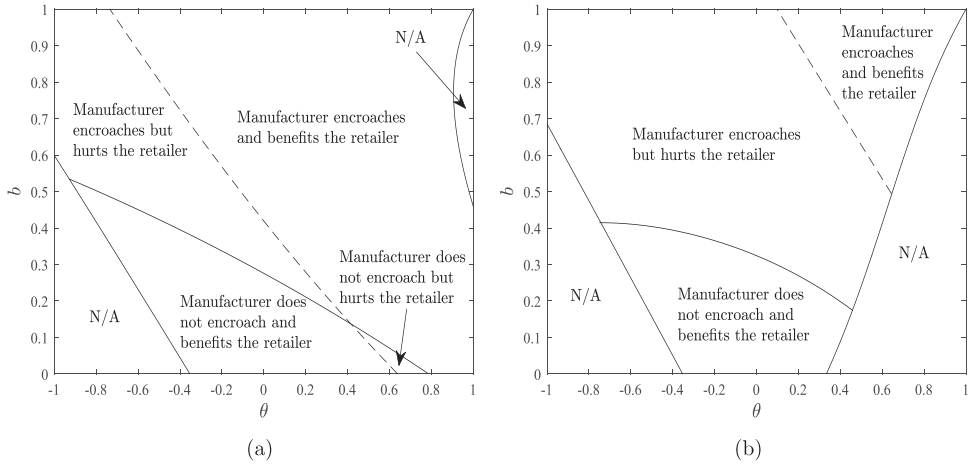


Fig. 4. Manufacturer decision and impact on the retailer with live streaming ($a=40$). (a) $m = 0.15$. (b) $m = 0.85$.

subsection, we examine the impact of the market share of the direct channel on the manufacturer's and retailer's strategies by assuming $m \in (0, 1)$, a larger m implies a larger market share of the direct channel under manufacturer encroachment. Therefore, we recalculate and get the new equilibrium results of case EW and case EL using the same method as in the basic model.

First, we explore manufacturer encroachment strategy and its impact on retailer's profit without and with live streaming, summarized as Proposition 8 and Fig. 4, respectively.

Proposition 8: Given that the retailer does not introduce live streaming.

i) The manufacturer encroaches if $0 < m < \frac{2}{3}$ and $\hat{b}_1 < b < 1$, or $\frac{2}{3} < m < 1$; otherwise, the manufacturer does not encroach.

ii) Manufacturer encroachment benefits the retailer if and only if $0 < m < \frac{1}{4}$ and $\frac{2\sqrt{2\,(3-4\,m)m}}{3-4\,m} < b < 1$.

A direct comparison between Propositions 8 (i) and 1 in the basic model reveals an important finding. In addition to the competition intensity (i.e., b), the market share of the direct channel (i.e., m) also affects the manufacturer encroachment strategy when the retailer does not introduce live streaming. Specifically, when the market share of the direct channel is small (i.e., $0 < m < \frac{2}{3}$), the manufacturer prefers encroachment only when the competition intensity is large (i.e., $\hat{b}_1 < b < 1$), for the reason similar to Proposition 1. However, when the market share of the direct channel is large (i.e., $\frac{2}{3} < m < 1$), the manufacturer could shift the main profit source to the direct channel through encroachment and make an independent decision on the direct price, which grants the manufacturer a higher dominant position, and therefore, the manufacturer always prefers encroachment. Furthermore, different from Proposition 4, when the market share of the direct channel is small (i.e., $0 < m < \frac{1}{4}$) and the channel competition is intense (i.e., $\frac{2\sqrt{2\,(3-4\,m)m}}{3-4\,m} < b < 1$), the retailer can benefit from manufacturer encroachment, as Proposition 8 (ii). The retailer benefits from the fact that the manufacturer has less incentive to raise wholesale price to shift demand to the direct channel and from the fact that intense competition mitigates the double marginalization of the reselling channel.

However, when the retailer introduces live streaming, the manufacturer's encroachment strategy and its impact on the retailer remain qualitatively unchanged when considering the market share of the direct channel m, as Fig. 4 shows.

Comparing the results revealed in Fig. 4(a) and (b), the magnitude of m will not qualitatively change the impact of θ and b on the manufacturer encroachment strategy. However, as m increases, the likelihood of manufacturer encroachment increases, while the likelihood of the retailer benefiting from manufacturer encroachment decreases. Intuitively, the larger market share the manufacturer competes for after encroachment makes him/her choose to encroach even in the face of a lower competitive intensity or a smaller spillover factors. However, the retailer can only profit from the lower wholesale price resulting from manufacturer encroachment when the competition intensity is more intense or when the flow effect of live streaming is more significant.

Next, Fig. 5 explores the retailer's live streaming strategy and its impact on the manufacturer. Compared to Fig. 3, we find that the main findings qualitatively remain essentially consistent with the basic model. Whether the retailer introduces live streaming remains closely related to the fixed cost (i.e., c_r): as c_r increases, the "EW" region widens and the "EL" region narrows. However, further research shows that the retailer's live streaming strategy is also influenced by the market share of the direct channel (i.e., m): as m increases, the "EW" region widens and the "EL" region narrows. The manufacturer's shift in sales focus to the direct channel brings higher wholesale price pressure, making the retailer more inclined not to introduce live streaming. Notably, the region where the manufacturer prefers the retailer not to introduce live streaming expands as well. In other words, a higher market share of the direct channel undermines the advantage of retailer live streaming. When the competition intensity is high and the spillover factor is small, a larger m makes it more likely that the retailer's live streaming behavior results in a "lose–lose" situation for both parties. In addition, as m

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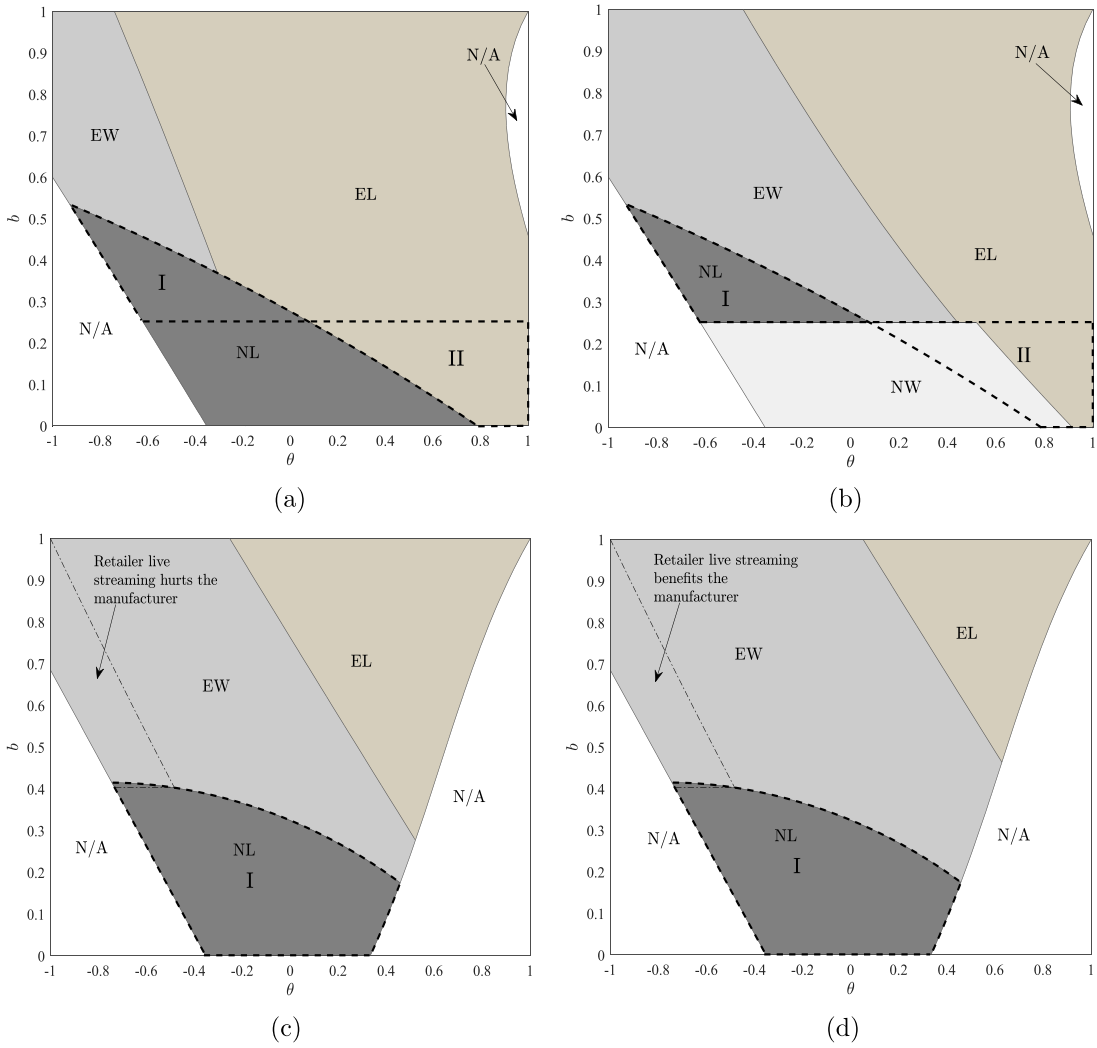


Fig. 5. Retailer decision and impact on the manufacturer ($a = 40$). (a) $m = 0.15, c = 70$. (b) $m = 0.15, c = 170$. (c) $m = 0.85, c = 70$. (d) $m = 0.85$, $c = 170$.

increases, the region where the retailer has to make a tradeoff between "EL" and "NW" shrinks and may eventually disappear. This is because, when competition intensity is small and the flow spillover is significant, a larger m means the manufacturer is more likely to encroach, regardless of whether the retailer introduces live streaming, thereby live streaming becomes less effective in promoting manufacturer encroachment. However, based on region I in Fig. 5, we can observe that retailer live streaming still may discourage manufacturer encroachment even when m is large. Therefore, the manufacturer and the retailer can reach the "NL" strategy if m is large.

B. Retailer Live Streaming After Manufacturer Encroachment

In the basic model, we assume that the retailer decides whether to introduce live streaming before manufacturer encroachment, to focus on the impact of retailer live streaming on manufacturer encroachment. In this section, we examine an alternative scenario in which the retailer determines whether to introduce live streaming after the manufacturer encroachment to further explore the interplay between manufacturer encroachment and retailer live streaming. We draw Fig. 6 to verify the robustness of the main findings and to observe the importance of the timing of the retailer's live streaming. It is clear that when the retailer introduces live streaming, she could profit from manufacturer encroachment if the spillover factor is extremely large. This is consistent with the findings of the basic model, and independent of the timing of the retailer's introduction of live streaming.

Furthermore, we find that when the retailer's live streaming decision is made after the manufacturer encroachment, it indeed receives the influence of the manufacturer encroachment. Specifically, in Fig. 6(a), when the fixed cost of introducing live streaming is low, the manufacturer faces a tradeoff between "EW" and "NL" when the spillover factor is small. When the competition intensity is large, manufacturer encroachment prevents the retailer from live streaming but hurts the retailer. In other words, the retailer prefers the "NL" equilibrium, i.e., the introduction of live streaming. When the fixed cost of live streaming is high, as illustrated in Fig. 6(b), the manufacturer faces a tradeoff between "EL" and "NW" when the spillover

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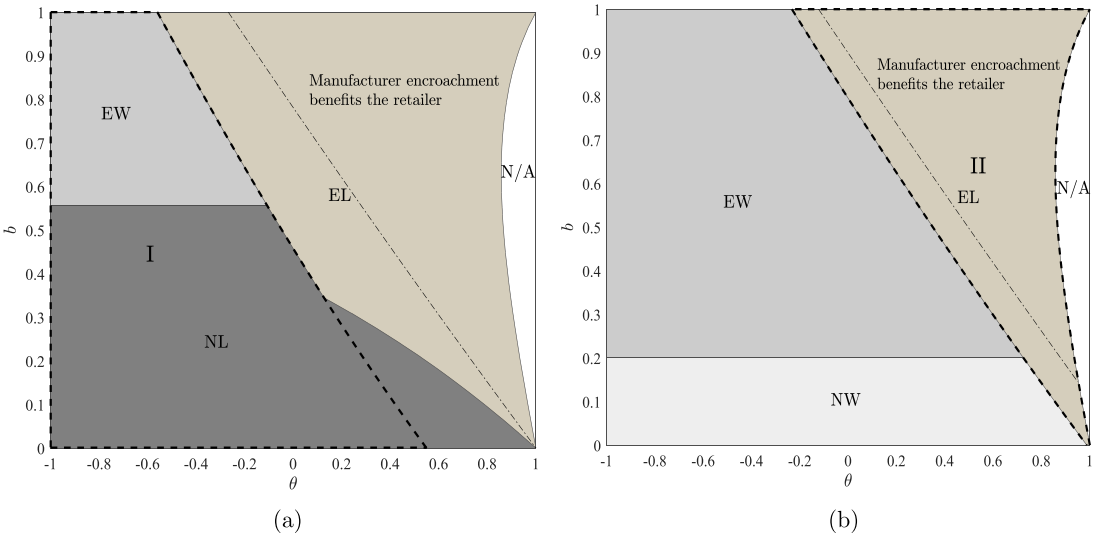


Fig. 6. Manufacturer decision and impact on retailer profit with live streaming ($a = 40$). (a) $c_r = 70$. (b) $c_r = 170$.

factor is large. Regardless of the competition intensity, the manufacturer always encroaches and incentivizes the retailer to introduce live streaming, which is more favorable to the manufacturer, and thus the equilibrium "EL" always occurs. However, manufacturer encroachment harms the retailer's revenue if the spillover factor is not significantly large. At this point, both parties are unable to realize a "win–win" situation in "EL". Recall that in the basic model, when the retailer decides before manufacturer encroachment, both parties can always achieve a "win–win" situation when the equilibrium strategy is "EL", even if the flow effect is not significantly large. Overall, the retailer's delay in the live streaming decision not only fails to change the manufacturer encroachment decision but also leads to a loss of initiative and is even forced to introduce live streaming under certain conditions, which reduces the value of the retailer's live streaming.

VI. Conclusion

Live streaming is increasingly popular in e-commerce retailing, which not only proves to be an effective means of improving sales but also serves as a strategy to affect manufacturer encroachment. In the face of manufacturer encroachment, it is crucial for retailers to determine whether to introduce live streaming. In this regard, we developed a theoretical model to investigate a retailer's live streaming strategy with a manufacturer who may encroach through a direct channel. Through a series of calculations and analyses, our conclusions enriched the existing studies on manufacturer encroachment and contributed theoretical foundations for manufacturers' channel expansion and retailers' live streaming introduction decisions. Several main findings obtained in this article are summarized as follows.

First, the retailer's live streaming strategy impacted the manufacturer's encroachment behavior and benefited the retailer from the manufacturer's encroachment. Specifically, when the retailer did not introduce live streaming, the manufacturer would only encroach when channel competition was fierce, which always hurt the retailer. However, when the retailer introduced live streaming, the manufacturer would choose to encroach either when channel competition was fierce or when the flow effect of live streaming was significant, and the manufacturer encroachment benefited the retailer when the flow effect was significant. Second, when the competition intensity was moderate and the flow effect of live streaming was insignificant, with the manufacturer always encroaching, introducing live streaming always hurted the retailer; otherwise, the retailer introduced live streaming when the fixed cost of live streaming was low and not when the fixed cost was high. Third, the retailer's live streaming could discourage or incentivize manufacturer encroachment, and the manufacturer might profit from the retailer's live streaming, even if it discouraged encroachment. Finally, we also considered several extensions to verify the robustness of our main models.

Our analysis also sheds light on several managerial implications, which was helpful for manufacturers and retailers implementing encroachment and live streaming strategies. First and foremost, when developing an encroachment or live streaming strategy, both manufacturers and retailers need to combine product characteristics with full consideration of factors such as channel substitutability and the type of spillover effects of live streaming. Second, retailers should plan their live streaming strategies as early as possible, including but not limited to the construction of live streaming teams and the design of live streaming programs to discourage manufacturers from encroaching. This suggestion is virtually consistent with the practice because we observed that the well-known manufacturer Procter & Gamble platform store in JD.com had opened live streaming sales, apparently under the significant cannibalization effect of JD.com live streaming, Procter & Gamble's product manufacturer Tide, which only had a medium competition intensity, had so far failed to open an official direct channel. AMD, a well-known semiconductor manufacturer in the United States,

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was also in this situation. This also suggested that manufacturers' encroachment strategies must be carefully formulated and that a single-minded attempt to open up official direct channels may be detrimental to their benefit instead. Third, reducing the fixed costs of live streaming gave retailers greater initiative and flexibility in responding to possible encroachment decisions by manufacturers, thereby retailers should reduce the fixed costs of building live rooms. For example, JD.com's "purchase and sell" live room was remodeled with a temporary conference room.

However, our study had some limitations. We only considered a single retailer and a single manufacturer in the supply chain and did not investigate scenarios involving multiple manufacturers and/or multiple retailers. In our study, we focused on retailers conducting live streaming independently. However, in reality, there were also cases where retailers collaborated with prominent live streamers or outsource live streaming to professional agencies. Additionally, exploring the manufacturer's potential introduction of live streaming is a promising direction for the future. Addressing these limitations will be the focus of our future research endeavors.

Appendix A
Proof of the Equilibrium Results

Case NW: No Encroachment and Without Live Streaming

When the retailer does not introduce live streaming, and the manufacturer chooses not to encroach, there is only the reselling channel in the supply chain, where the manufacturer wholesales the product to the retailer at a wholesale price w. The retailer then sells the product to consumers at a retail price p_r. In this case, the manufacturer first decides on the wholesale price w, followed by the retailer's decision on the retail price p_r. We will use backward induction to solve for the optimal outcome. First, the retailer maximizes her profit by choosing the optimal p_r:

$$\max_{p_r} \ \pi_r = (p_r - w)(a - p_r). \tag{7}$$

By solving (7), we can obtain the retailer's optimal decision in the second stage, which is: $p_r = \frac{a+w}{2}$. Based on the retailer's response, the manufacturer decides the optimal wholesale price w to maximize his profit:

$$\max_{w} \ \pi_m = w(a - p_r). \tag{8}$$

Therefore, by solving (8), we can obtain the manufacturer's optimal decision $w^{\mathrm{NW}} = \frac{a}{2}$. At this point, by substituting w into the response function for p_r, and the profit functions, we can obtain the equilibrium decision $p_r^{\mathrm{NW}} = \frac{3a}{4}$, the optimal profits $\pi_m^{\mathrm{NW}} = \frac{a^2}{8}$, $\pi_r^{\mathrm{NW}} = \frac{a^2}{16}$.

Case EW: Encroachment But Without Live Streaming

When the retailer does not introduce live streaming and the manufacturer chooses to encroach, the supply chain consists of the direct channel and the reselling channel. In the indirect sales channel, the manufacturer wholesales the product to the retailer at price w, and the retailer sells the product to consumers at the retail price p_r. In the direct channel, the manufacturer directly sells the product to consumers at the retail price p_m. In this case, the manufacturer first determines the wholesale price w, and then both the manufacturer and the retailer simultaneously decide the retail prices p_m and p_r. We continue to use backward induction to determine the optimal states in this strategy combination. First, we calculate the optimal price decisions of the manufacturer and the retailer in the second stage:

$$\begin{cases} \max\limits_{p_m} \ \pi_m = p_m(\frac{a}{2} - p_m + bp_r) + w(\frac{a}{2} - p_r + bp_m) \\ \max\limits_{p_r} \ \pi_r = (p_r - w)(\frac{a}{2} - p_r + bp_m). \end{cases} \tag{9}$$

By solving (9), we can obtain the optimal price decisions of the manufacturer and the retailer in the second stage, which are: $p_m = \frac{a(b+2)+6bw}{8-2b^2}$ and $p_r = \frac{a(b+2)+2(b^2+2)w}{8-2b^2}$.

Based on the optimal responses, the manufacturer decides the optimal wholesale price w to maximize his profit:

$$\max_{w} \ \pi_m = p_m\left(\frac{a}{2} - p_m + bp_r\right) + w\left(\frac{a}{2} - p_r + bp_m\right). \tag{10}$$

Therefore, by solving (10), we can obtain the optimal decision of the manufacturer in the first stage: $w^{\mathrm{EW}} = \frac{a(b^3+8)}{4(1-b)(b^2+8)}$. At this point, by substituting w into response functions for p_m and p_r, and the profit functions, we can obtain the equilibrium decisions $p_m^{\mathrm{EW}} = \frac{a(4-b)(b+2)}{4(1-b)(b^2+8)}$, $p_r^{\mathrm{EW}} = \frac{a(b^3-2b^2+4b-12)}{4(b-1)(b^2+8)}$, the optimal profits $\pi_m^{\mathrm{EW}} = \frac{a^2(b^3+b^2+4b+12)}{16(1-b)(b^2+8)}$, $\pi_r^{\mathrm{EW}} = \frac{a^2(b^2+2)^2}{4(b^2+8)^2}$.

Case NL: No Encroachment but Live Streaming

When the retailer introduces live streaming and the manufacturer chooses not to encroach, there is only the reselling channel in the supply chain. In this case, the manufacturer first decides the wholesale price w, then the retailer decides the retail price p_r, and finally, the retailer determines the live streaming effort level e. Applying backward induction, the retailer maximizes her profit by determining the live streaming effort level e:

$$\max_{e} \ \pi_r = (p_r - w)(a - p_r + e) - c_r - c_l. \tag{11}$$

By solving (11), we can obtain the retailer's optimal decision $e = p_r - w$. Then, the retailer decides the optimal retail price $p_r = a$ in the same way. Based on the retailer's optimal responses, the manufacturer determines the wholesale price by solving the following function:

$$\max_{w} \ \pi_m = w(a - p_r + e). \tag{12}$$

Therefore, by solving (12), we can obtain the manufacturer's optimal decision $w^{\mathrm{NL}} = \frac{a}{2}$. At this point, by substituting w into the response functions for p_r and e, and the profit functions, we can obtain the equilibrium decisions $e^{\mathrm{NL}} = \frac{a}{2}$, $p_r^{\mathrm{NL}} = a$, the the optimal profits $\pi_m^{\mathrm{NL}} = \frac{a^2}{4}$, $\pi_r^{\mathrm{NL}} = \frac{a^2}{8} - c_r$.

Case EL: Encroachment and Live Streaming

When the retailer introduces live streaming and the manufacturer chooses to encroach, there are two channels in the supply chain: direct and reselling channels. In the direct channel, the manufacturer sells products directly to consumers at the direct price p_m. In the reselling channel, the manufacturer collaborates with the retailer to sell products through wholesale. Additionally, the retailer needs to determine the appropriate live streaming effort level. The retailer incurs both fixed cost c_r and variable cost c_l. Using backward induction, we first calculate the optimal decision for the retailer in the third stage. In this stage, the retailer maximizes her profit by determining the live streaming effort

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level e:

$$\max_{e}\ \pi_r = (p_r - w)\left(\frac{a}{2} - p_r + bp_m + e\right) - c_r - c_l. \quad (13)$$

By solving (13), we can obtain the optimal decision for the retailer $e = p_r - w$. Based on the retailer's optimal response, both the retailer and the manufacturer decide the product prices, p_m and p_r, respectively, by maximize their profit:

$$\begin{cases} \max\limits_{p_m}\ \pi_m = p_m(\frac{a}{2} - p_m + bp_r + \theta e) + w(\frac{a}{2} - p_r + bp_m + e) \\ \max\limits_{p_r}\ \pi_r = (p_r - w)(\frac{a}{2} - p_r + bp_m + e) - c_r - c_l. \end{cases} \quad (14)$$

By solving (14), we can obtain the optimal decisions for the manufacturer and the retailer $p_m = \frac{ab+a\theta+2\,bw-2\,\theta\,w+a}{2(2-b^2-b\theta)}$, $p_r = \frac{2\,b^2w-2\,b\theta\,w+ab+2\,a}{2(2-b^2-b\theta)}$. Then the manufacturer decides the wholesale price w by solving the following function:

$$\max_{w}\ \pi_m = p_m\left(\frac{a}{2} - p_m + bp_r + \theta e\right) + w\left(\frac{a}{2} - p_r + bp_m + e\right). \quad (15)$$

The optimal decision of the manufacturer $w^{\mathrm{EL}} = \frac{a(b^4+2\,b^3\theta+b^2\theta^2-2\,b^2-4\,b\theta-2\,\theta^2+2\,b-2\,\theta+4)}{4(b-1)(b+1)(b+\theta+2)(b+\theta-2)}$. By substituting w into the response functions for p_m, p_r, and e, and the profit functions, we can obtain the equilibrium decisions $e^{\mathrm{EL}} = \frac{a(b^2+b\theta+b+\theta+2)}{2(b+\theta+2)(2-b-\theta)}$, $p_m^{\mathrm{EL}} = \frac{a(3\,b+\theta+2)(b^2+b\theta-2)}{4(1-b)(b+1)(b+\theta+2)(b+\theta-2)}$, $p_r^{\mathrm{EL}} = \frac{a(b^4-b^2\theta^2+2\,b^3+2\,b^2\theta+4\,b^2+2\,b\theta+2\,\theta^2-4\,b-8)}{4(1-b)(b+1)(b+\theta+2)(b+\theta-2)}$, the optimal profits $\pi_m^{\mathrm{EL}} = \frac{a^2(b^4+2\,b^3\theta+b^2\theta^2+4\,b^2+4\,b\theta+12\,b+4\,\theta+8)}{16\,(b-1)(b+1)(b+\theta+2)(b+\theta-2)}$, $\pi_r^{\mathrm{EL}} = \frac{a^2(b^2+b\theta+b+\theta+2)^2}{8\,(b^2+2\,b\theta+\theta^2-4)^2} - c_r$.

To ensure that all equilibrium outcomes are positive, as well as to ensure that the objective functions with respect to decision variables remain concave in all four cases above, we have $-1 < \theta < \hat{\theta}\,(0 < \hat{\theta} < 1)$, where $\hat{\theta} = \frac{b^3+\sqrt{-4\,b^3-4\,b^2+8\,b+9}-2\,b-1}{2-b^2}$.

Appendix B
Proof of the Propositions

Proof of Proposition 1

Depending on the equilibrium results under different cases, we have: $\pi_m^{\mathrm{NW}} = \frac{a^2}{8}$; $\pi_m^{\mathrm{EW}} = -\frac{a^2(b^3+b^2+4b+12)}{16(b-1)(b^2+8)}$. So, $\pi_m^{\mathrm{EW}} - \pi_m^{\mathrm{NW}} = \frac{a^2(3b^3-b^2+20b-4)}{16(1-b)(b^2+8)}$. So, we can get that the size relation between π_m^{EW} and π_m^{NW} depends on $3b^3 - b^2 + 20b - 4$. We assume that the function $f(b) = 3b^3 - b^2 + 20b - 4$. We obtain that b needs to satisfy $\frac{1}{9}(1 - \frac{179}{\sqrt[3]{217+54\sqrt{1983}}} + \sqrt[3]{217+54\sqrt{1983}}) < b < 1$ when $f_1(b) > 0$ and $0 < b < \frac{1}{9}(1 - \frac{179}{\sqrt[3]{217+54\sqrt{1983}}} + \sqrt[3]{217+54\sqrt{1983}})$ when $f_1(b) < 0$. We set $b_1 = \frac{1}{9}(1 - \frac{179}{\sqrt[3]{217+54\sqrt{1983}}} + \sqrt[3]{217+54\sqrt{1983}})$. So, we can get $\pi_m^{\mathrm{EW}} > \pi_m^{\mathrm{NW}}$ when $b_1 < b < 1$; otherwise, $\pi_m^{\mathrm{EW}} < \pi_m^{\mathrm{NW}}$.

Proof of Proposition 2

Given that the retailer introduces live streaming, we compare the retailer's live streaming effort level e with and without manufacturer encroachment. We have: $e^{\mathrm{NL}} = \frac{a}{2}$, $e^{\mathrm{EL}} = \frac{a(b^2+b\theta+b+\theta+2)}{2(b+\theta+2)(2-b-\theta)}$. So, $e^{\mathrm{EL}} - e^{\mathrm{NL}} = -\frac{a(2\,b^2+3\,b\theta+\theta^2+b+\theta-2)}{2\,(b+\theta+2)(b+\theta-2)}$. Therefore, we can obtain one root $\theta_1 = -\frac{3}{2}\,b - \frac{1}{2} + \frac{1}{2}\sqrt{b^2+2\,b+9}$ of $f_1(\theta) = 2\,b^2 + 3\,b\theta + \theta^2 + b + \theta - 2$, and another root is not viable (i.e., $-\frac{3}{2}\,b - \frac{1}{2} - \frac{1}{2}\sqrt{b^2+2\,b+9} < -1$). So, we can get $e^{\mathrm{EL}} > e^{\mathrm{NL}}$ when $\theta_1 < \theta < \hat{\theta}$; otherwise, $e^{\mathrm{EL}} < e^{\mathrm{NL}}$.

Proof of Proposition 3

Depending on the equilibrium results under different combinations of strategies, we have: $\pi_m^{\mathrm{NL}} = \frac{a^2}{4}$; $\pi_m^{\mathrm{EL}} = \frac{a^2(b^4+2\,b^3\theta+b^2\theta^2+4\,b^2+4\,b\theta+12\,b+4\,\theta+8)}{16\,(b-1)(b+1)(b+\theta+2)(b+\theta-2)}$. So, $\pi_m^{\mathrm{EL}} - \pi_m^{\mathrm{NL}} = \frac{a^2(3\,b^4+6\,b^3\theta+3\,b^2\theta^2-24\,b^2-12\,b\theta-4\,\theta^2-12\,b-4\,\theta+8)}{16(1-b)(b+1)(b+\theta+2)(b+\theta-2)}$. Therefore, we can obtain one root $\theta_2 = \frac{3\,b^3+2\sqrt{12\,b^4+6\,b^3-21\,b^2-6\,b+9}-6\,b-2}{4-3\,b^2}$ of $f_2(\theta) = 3\,b^4 + 6\,b^3\theta + 3\,b^2\theta^2 - 24b^2 - 12b\theta - 4\theta^2 - 12b - 4\theta + 8$, and another root is always not viable (i.e., $\frac{3\,b^3-2\sqrt{12\,b^4+6\,b^3-21\,b^2-6\,b+9}-6\,b-2}{4-3\,b^2} < -1$). While $\theta_2 \in (-1, \hat{\theta})$ only when $0 < b < b_2$, where b_2 is the only root of $3b^4 - 6b^3 - 21b^2 + 8$ that satisfies $0 < b < 1$. So, we can get $\pi_m^{\mathrm{EL}} < \pi_m^{\mathrm{NL}}$ when $-1 < \theta < \theta_2$ and $0 < b < b_2$; otherwise, $\pi_m^{\mathrm{EL}} > \pi_m^{\mathrm{NL}}$.

Proof of Proposition 4

Depending on the equilibrium results in the four cases, we have: $\pi_r^{\mathrm{NW}} = \frac{a^2}{16}$; $\pi_r^{\mathrm{EW}} = \frac{a^2(b^2+2)^2}{4(b^2+8)^2}$; $\pi_r^{\mathrm{NL}} = \frac{a^2}{8} - c_r$; $\pi_r^{\mathrm{EL}} = \frac{a^2(b^2+b\theta+b+\theta+2)^2}{8\,(b^2+2\,b\theta+\theta^2-4)^2} - c_r$. In the absence of live streaming, $\pi_r^{\mathrm{EW}} - \pi_r^{\mathrm{NW}} = \frac{3\,a^2(b-2)(b+2)(b^2+4)}{16\,(b^2+8)^2} < 0$. So, we can get $\pi_r^{\mathrm{EW}} < \pi_r^{\mathrm{NW}}$ always holds. In the presence of live streaming, $\pi_r^{\mathrm{EL}} - \pi_r^{\mathrm{NL}} = -\frac{a^2(b\theta+\theta^2-b-\theta-6)(2\,b^2+3\,b\theta+\theta^2+b+\theta-2)}{8\,(b+\theta+2)^2(b+\theta-2)^2}$. We can obtain $\theta_1 = -\frac{3}{2}\,b - \frac{1}{2} + \frac{1}{2}\sqrt{b^2+2\,b+9}$ of $f_1(\theta) = 2\,b^2 + 3\,b\theta + \theta^2 + b + \theta - 2$, and another root is not viable (i.e., $-\frac{3}{2}\,b - \frac{1}{2} - \frac{1}{2}\sqrt{b^2+2\,b+9} < -1$). Therefore, we can get $\pi_r^{\mathrm{EL}} < \pi_r^{\mathrm{NL}}$ when $-1 < \theta < \theta_1$; otherwise, $\pi_r^{\mathrm{EL}} > \pi_r^{\mathrm{NL}}$.

Proof of Proposition 5

Depending on the equilibrium results under different combinations of strategies, we have: $\pi_r^{\mathrm{NW}} = \frac{a^2}{16}$; $\pi_r^{\mathrm{NL}} = \frac{a^2}{8} - c_r$; $\pi_r^{\mathrm{EW}} = \frac{a^2(b^2+2)^2}{4(b^2+8)^2}$; $\pi_r^{\mathrm{EL}} = \frac{a^2(b^2+b\theta+b+\theta+2)^2}{8\,(b^2+2\,b\theta+\theta^2-4)^2} - c_r$.

So, according to the manufacturer's encroachment strategy under different live streaming strategies given in Propositions 1 and 3, we first obtain $b_1 < b_2$, and then we make the following comparisons to determine the retailer's live streaming strategy.

i) If $0 < b < b_1$ and $-1 < \theta < \theta_2$, $\pi_r^{\mathrm{NL}} - \pi_r^{\mathrm{NW}} = \frac{a^2}{16} - c_r$, there exists $c_{r1} = \frac{a^2}{16}$, when $c < c_{r1}$, $\pi_r^{\mathrm{NL}} > \pi_r^{\mathrm{NW}}$; otherwise, $\pi_r^{\mathrm{NL}} < \pi_r^{\mathrm{NW}}$.

ii) If $0 < b < b_1$ and $\theta_2 < \theta < \hat{\theta}$, $\pi_r^{\mathrm{EL}} - \pi_r^{\mathrm{NW}} = \frac{a^2(b^2+b\theta+b+\theta+2)^2}{8\,(b^2+2\,b\theta+\theta^2-4)^2} - \frac{a^2}{16} - c_r$, there exists $c_{r2} = \frac{a^2(b^4-4\,b^2\theta^2-4\,b\theta^3-\theta^4+4\,b^3+8\,b^2\theta+4\,b\theta^2+18\,b^2+28\,b\theta+10\,\theta^2+8\,b+8\,\theta-8)}{16\,(b+\theta+2)^2(b+\theta-2)^2}$, where $c_{r2} > 0$ always holds when $0 < b < b_1$ and $\theta_2 < \theta < \hat{\theta}$. Therefore, when $c < c_{r2}$, $\pi_r^{\mathrm{EL}} > \pi_r^{\mathrm{NW}}$; otherwise, $\pi_r^{\mathrm{EL}} < \pi_r^{\mathrm{NW}}$.

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iii) If $b_1 < b < b_2$ and $-1 < \theta < \theta_2$, $\pi_r^{\mathrm{NL}} - \pi_r^{\mathrm{EW}} = \frac{a^2(-b^4+8b^2+56)}{8(b^2+8)^2} - c_r$, there exists $c_{r3} = \frac{a^2(-b^4+8b^2+56)}{8(b^2+8)^2}$, when $c < c_{r3}$, $\pi_r^{\mathrm{NL}} > \pi_r^{\mathrm{EW}}$; otherwise, $\pi_r^{\mathrm{NL}} < \pi_r^{\mathrm{EW}}$.

iv) If $b_1 < b < b_2$ and $\theta_2 < \theta < \hat{\theta}$, or $b_2 < b < 1$, $\pi_r^{\mathrm{EL}} - \pi_r^{\mathrm{EW}} = \frac{a^2(b^2+b\theta+b+\theta+2)^2}{8(b^2+2b\theta+\theta^2-4)^2} - \frac{a^2(b^2+2)^2}{4(b^2+8)^2} - c_r$, we can get $c_{r4} < 0$ when $b_3 < b < b_2$ and $\theta_2 < \theta < \theta_3$, or $b_2 < b < b_4$ and $-1 < \theta < \theta_3$, where θ_3 is the only root of $f_3(\theta) = 2(b^2+2)^2\theta^4 + 8b(b^2+2)^2\theta^3 + (11b^6 - 2b^5 + 15b^4 - 32b^3 - 96b^2 - 128b - 128)\theta^2 + (6b^7 - 4b^6 - 38b^5 - 68b^4 - 320b^3 - 320b^2 - 512b - 256)\theta + b^8 - 2b^7 - 29b^6 - 36b^5 - 172b^4 - 192b^3 - 320b^2 - 256b - 128$ because $f_3(\theta)$ is decreasing with respect to θ, b_3 is the only solution of $f_3(\theta = \theta_2) = 0$ that satisfies $b_1 < b < b_2$, b_4 is the only solution of $f_3(\theta = -1) = 0$ (i.e., $b^8 - 8b^7 - 14b^6 - 8b^5 - 87b^4 + 64b^3 - 88b^2 + 96b + 8 = 0$) that satisfies $b_2 < b < 1$; otherwise, $c_{r4} > 0$. Therefore, when $b_3 < b < b_2$ and $\theta_2 < \theta < \theta_3$, or $b_2 < b < b_4$ and $-1 < \theta < \theta_3$, $\pi_r^{\mathrm{EL}} < \pi_r^{\mathrm{EW}}$; otherwise, $\pi_r^{\mathrm{EL}} < \pi_r^{\mathrm{EW}}$ when $c > c_{r4}$ and $\pi_r^{\mathrm{EL}} > \pi_r^{\mathrm{EW}}$ when $c < c_{r4}$.

In summary, we can get when $b_3 < b < b_2$ and $\theta_2 < \theta < \theta_3$, or $b_2 < b < b_4$ and $-1 < \theta < \theta_3$, $\pi_r^{\mathrm{L}} < \pi_r^{\mathrm{W}}$ always holds; otherwise, there exists and only exists a threshold on $c_{ri}, i \in \{1,2,3,4\}$ in each threshold range such that $\pi_r^{\mathrm{L}} < \pi_r^{\mathrm{W}}$ when $c > c_{ri}$ and $\pi_r^{\mathrm{L}} > \pi_r^{\mathrm{W}}$ when $c < c_{ri}$.

Proof of Proposition 6

As Propositions 1 and 3 shown, when $0 < b < b_1$, the manufacturer does not encroach when the retailer does not introduce live streaming, while the manufacturer encroaches if $0 < b < b_2$ and $\theta_2 < \theta < \hat{\theta}$, or $b_2 < b < 1$ when the retailer introduces live streaming. According to Proposition 5, we can obtain $b_1 < b_2$ and when $0 < b < b_1$ and $\theta_2 < \theta < \hat{\theta}$, the retailer introduces live streaming when $c < c_{r2}$, which prompts the manufacturer to encroach, at which point the equilibrium shifts from NW to EL.

Similarly, when $b_1 < b < 1$, the manufacturer encroaches when the retailer does not introduce live streaming, while the manufacturer does not encroach if $0 < b < b_2$ and $-1 < \theta < \theta_2$ when the retailer introduces live streaming. According to Proposition 5, we can obtain when $b_1 < b < b_2$ and $-1 < \theta < \theta_2$, the retailer introduces live streaming when $c < c_{r3}$, which discourages the manufacturer from encroaching, at which point the equilibrium shifts from EW to NL.

Proof of Proposition 7

Depending on the equilibrium results in the four cases, we have: $\pi_m^{\mathrm{NW}} = \frac{a^2}{8}$; $\pi_m^{\mathrm{EW}} = \frac{a^2(b^3+b^2+4b+12)}{16(1-b)(b^2+8)}$; $\pi_m^{\mathrm{NL}} = \frac{a^2}{4}$; $\pi_m^{\mathrm{EL}} = \frac{a^2(b^4+2b^3\theta+b^2\theta^2+4b^2+4b\theta+12b+4\theta+8)}{16(b-1)(b+1)(b+\theta+2)(b+\theta-2)}$.

So, according to the manufacturer's encroachment strategy under different live streaming strategies given in Propositions 1 and 3, we first obtain $b_1 < b_2$, and then we make the following comparisons to explore the impact of the retailer's live streaming strategy on the manufacturer.

i) If $0 < b < b_1$ and $-1 < \theta < \theta_2$, $\pi_m^{\mathrm{NL}} - \pi_m^{\mathrm{NW}} = \frac{a^2}{8} > 0$. Therefore, $\pi_m^{\mathrm{NL}} > \pi_m^{\mathrm{NW}}$ always holds.

ii) If $0 < b < b_1$ and $\theta_2 < \theta < \hat{\theta}$, $\pi_m^{\mathrm{EL}} - \pi_m^{\mathrm{NW}} = \frac{a^2(b^4+2b^3\theta+b^2\theta^2-14b^2-8b\theta-2\theta^2-12b-4\theta)}{16(1-b)(b+1)(b+\theta+2)(b+\theta-2)}$, where $b^4 + 2b^3\theta + b^2\theta^2 - 14b^2 - 8b\theta - 2\theta^2 - 12b - 4\theta > 0$ always holds when $0 < b < b_1$ and $\theta_2 < \theta < \hat{\theta}$. Therefore, $\pi_m^{\mathrm{EL}} > \pi_m^{\mathrm{NW}}$ always holds.

iii) If $b_1 < b < b_2$ and $-1 < \theta < \theta_2$, $\pi_m^{\mathrm{NL}} - \pi_m^{\mathrm{EW}} = \frac{a^2(5b^3-3b^2+36b-20)}{16(b-1)(b^2+8)}$. Therefore, there exists b_5 is the only root of $5b^3 - 3b^2 + 36b - 20$ that satisfies $b_1 < b < b_2$. So, we can get $\pi_m^{\mathrm{NL}} < \pi_m^{\mathrm{EW}}$ when $b_5 < b < b_2$ and $-1 < \theta < \theta_2$; otherwise, $\pi_m^{\mathrm{NL}} > \pi_m^{\mathrm{EW}}$.

iv) If $b_1 < b < b_2$ and $\theta_2 < \theta < \hat{\theta}$, or $b_2 < b < 1$, $\pi_m^{\mathrm{EL}} - \pi_m^{\mathrm{EW}} = \frac{a^2(b^4+2b^3\theta+b^2\theta^2+4b^2+4b\theta+12b+4\theta+8)}{16(b-1)(b+1)(b+\theta+2)(b+\theta-2)} - \frac{a^2(b^3+b^2+4b+12)}{16(1-b)(b^2+8)}$. There exists one root $\theta_4 = \frac{-2b^5-2b^4-15b^3-18b^2+2\sqrt{2}\sqrt{(b+1)(b^3+6b+2)(b^2+2)^2}-28b-16}{2b^4+2b^3+13b^2+16b+12}$ of $f_4(\theta) = (2b^4 + 2b^3 + 13b^2 + 16b + 12)\theta^2 + (4b^5 + 4b^4 + 30b^3 + 36b^2 + 56b + 32)\theta + 2b^6 + 2b^5 + 13b^4 + 20b^3 + 32b^2 + 32b + 16$, and another root $\frac{-2b^5-2b^4-15b^3-18b^2-2\sqrt{2}\sqrt{(b+1)(b^3+6b+2)(b^2+2)^2}-28b-16}{2b^4+2b^3+13b^2+16b+12} < -1$, which is always viable. While θ_4 is viable when $b_5 < b < b_2$ and $\theta_2 < \theta < \hat{\theta}$, or $b_2 < b < 1$, where b_5 is the only root of $5b^3 - 3b^2 + 36b - 20$ that satisfies $b_1 < b < b_2$. Therefore, we can get $\pi_m^{\mathrm{EL}} < \pi_m^{\mathrm{EW}}$ when $b_5 < b < b_2$ and $\theta_2 < \theta < \theta_4$, or $b_2 < b < 1$ and $-1 < \theta < \theta_4$.

In summary, we can get $\pi_m^{\mathrm{L}} < \pi_m^{\mathrm{W}}$ when $b_5 < b < 1$ and $-1 < \theta < \theta_4$; otherwise, $\pi_m^{\mathrm{L}} > \pi_m^{\mathrm{W}}$.

Proof of Proposition 8

Following a same logic, we can derive the optimal solutions under manufacturer encroachment, i.e., case EW and case EL. We omit the straightforward algebraic step while only present the optimal solutions.

In case EW, we obtain the equilibrium decisions $w^{\mathrm{EW}*} = \frac{a(b^4m-b^4-b^3m-8bm+8m-8)}{2(b-1)(b+1)(b^2+8)}$, $p_r^{\mathrm{EW}*} = \frac{(b^4m-b^4+b^3m+2b^2m-2b^2+8bm-12m+12)a}{2(1-b)(b+1)(b^2+8)}$, $p_m^{\mathrm{EW}*} = \frac{(b^3m-b^3+b^2m-10bm+10b+8m)a}{2(1-b)(b+1)(b^2+8)}$, the optimal profits $\pi_m^{\mathrm{EW}*} = \frac{a^2((b-1)(b-2)(b^2+b+6)m^2-2(b-1)(b^3+4b-4)m+(b^2+2)^2)}{4(1-b)(b+1)(b^2+8)}$, $\pi_r^{\mathrm{EW}*} = \frac{a^2(b^2+2)^2(m-1)^2}{(b^2+8)^2}$.

In case EL, we obtain the equilibrium decisions, inline equation shown at the top of this page the optimal profits, inline equation shown at the top of this page .

In order to ensure equilibrium outcomes are positive and objective functions remain concave, we have, inline equation shown at the top of next page.

Inline equation shown at the top of next page we can get that the size relation between π_m^{NW} and $\pi_m^{\mathrm{EW}*}$ depends on $f^*(b) = (2m^2 - 4m + 3)b^4 - 4m(m-1)b^3 + (10m^2 - 16m + 15)b^2 - 32m(m-1)b + 8m(3m-2)$, which is incremental with respect to b in $b \in (0,1)$. and when $b = 0$,

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$$w^{\mathrm{EL}*}=\frac{a(b^4m+2\,b^3m\theta+b^2m\theta^2-b^4-2\,b^3\theta-b^2\theta^2-2\,b^2m-4\,bm\theta-2\,m\theta^2+2\,b^2-2\,bm+4\,b\theta+2\,m\theta+2\,\theta^2+4\,m-4)}{2\,(1-b)(b+1)(b+\theta+2)(b+\theta-2)},$$

$$p_r^{\mathrm{EL}*}=\frac{a(b^4m-b^2m\theta^2-b^4-2\,b^3m-2\,b^2m\theta+b^2\theta^2+4\,b^2m+2\,bm\theta+2\,m\theta^2-4\,b^2+4\,bm-2\,b\theta-2\,\theta^2-8\,m+8)}{2\,(b-1)(b+1)(b+\theta+2)(b+\theta-2)},$$

$$p_m^{\mathrm{EL}*}=\frac{a(b^2+b\theta-2)(3\,bm+m\theta-3\,b-2\,m-\theta)}{2\,(b-1)(b+1)(b+\theta+2)(b+\theta-2)},$$

$$e^{\mathrm{EW}*}=\frac{a(b^2m+bm\theta-b^2-bm-b\theta-m\theta+2\,m-2)}{(b+\theta+2)(b+\theta-2)},$$

$$\pi_m^{\mathrm{EL}*}=\frac{a^2((b^4+2\,b^3\theta+b^2\theta^2+4\,b^2+4\,b\theta-12\,b-4\,\theta+8)m^2+(-2\,b^4-4\,b^3\theta-2\,b^2\theta^2-8\,b^2-8\,b\theta+12\,b+4\,\theta-8)m+(b^2+b\theta+2)^2)}{4\,(b-1)(b+1)(b+\theta+2)(b+\theta-2)},$$

$$\pi_r^{\mathrm{EL}*}=\frac{a^2(b^2m+bm\theta-b^2-bm-b\theta-m\theta+2\,m-2)^2}{2\,(b+\theta+2)^2(b+\theta-2)^2}-c_r$$

$$\max\{\tfrac{3\,bm-3\,b-2\,m}{1-m},\tfrac{b^2m-b^2-bm+2\,m-2}{m+b-bm},\tfrac{b^2m-b^2+\sqrt{4\,bm^2-4\,bm+m^2-2\,m+1}+m-1}{(1-m)b},-1\}<\theta<\min\{\tfrac{-b^3m+b^3+2\,bm+\sqrt{4\,b^3m^2-4\,b^3m-4\,b^2m^2+8\,b^2m-8\,bm^2-4\,b^2+8\,bm+9\,m^2-16\,m+8}-2\,b-m}{b^2m-b^2-2\,m+2},1\}$$

$$\pi_m^{\mathrm{NW}}-\pi_m^{\mathrm{EW}*}=\frac{a^2((2\,m^2-4\,m+3)b^4-4\,m(m-1)b^3+(10\,m^2-16\,m+15)b^2-32\,m(m-1)b+8\,m(3\,m-2))}{8\,(b-1)(b+1)(b^2+8)},$$

$f^*(b=0)=8\,m(3\,m-2)$. Therefore, if $0<m<\frac{2}{3}$ there exists a unique root $\hat{b}_1$ of $f^*(b)$, and if $\frac{2}{3}<m<1$, there is always $f^*(b)>0$. Thus, if $0<m<\frac{2}{3}$ and $\hat{b}_1<b<1$, or $\frac{2}{3}<m<1$, $\pi_m^{\mathrm{NW}}-\pi_m^{\mathrm{EW}*}<0$ holds; otherwise, $\pi_m^{\mathrm{NW}}-\pi_m^{\mathrm{EW}*}>0$.

$\pi_r^{\mathrm{NW}}-\pi_r^{\mathrm{EW}*}=-\frac{a^2(4\,b^2m-3\,b^2+8\,m)(4\,b^2m-5\,b^2+8\,m-16)}{16\,(b^2+8)^2}$. It is easy to get $4\,b^2m-5\,b^2+8\,m-16<0$, so we can get that the size relation between π_r^{NW} and $\pi_r^{\mathrm{EW}*}$ depends on $4\,b^2m-3\,b^2+8\,m$. When $0<m<\frac{1}{4}$ and $\frac{2\sqrt{2\,(3-4\,m)m}}{3-4\,m}<b<1$, $4\,b^2m-3\,b^2+8\,m<0$, i.e., $\pi_r^{\mathrm{NW}}-\pi_r^{\mathrm{EW}*}<0$, otherwise, $\pi_r^{\mathrm{NW}}-\pi_r^{\mathrm{EW}*}>0$.

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